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(a)	$E = pc$
(b)(i)	$\Phi = 2.30 \times 1.60 \times 10^{-19}$ $= 3.68 \times 10^{-19}$ J
	$f_0 = \dfrac{\Phi}{h}$ $= \dfrac{3.68 \times 10^{-19}}{6.63 \times 10^{-34}}$ $= 5.55 \times 10^{14}$ Hz
(ii)	$E_n = \dfrac{h^2}{8mL^2} n^2$ $= \dfrac{\left(6.63 \times 10^{-34}\right)^2}{8\left(9.11 \times 10^{-31}\right)\left(3.0 \times 10^{-9}\right)^2} n^2$
	$\Delta E = \dfrac{\left(6.63 \times 10^{-34}\right)^2}{8\left(9.11 \times 10^{-31}\right)\left(3.0 \times 10^{-9}\right)^2}\left(8^2 - 2^2\right)$ $= 4.021 \times 10^{-19}$ J
	$\Delta E = \dfrac{4.021 \times 10^{-19}}{1.60 \times 10^{-19}}$ $= 2.513$ eV $= 2.51$ eV
(iii)	$E_k = 2.51 - 2.30$ $= 0.21$ eV
	$E_k = \dfrac{p^2}{2m}$ $p = \sqrt{2mE_k}$ $= \sqrt{2\left(9.11 \times 10^{-31}\right)\left(0.21 \times 1.60 \times 10^{-19}\right)}$ $= 2.4743 \times 10^{-25}$ kg m s^{-1}
	$\lambda = \dfrac{h}{p}$ $= \dfrac{6.63 \times 10^{-34}}{2.4743 \times 10^{-25}}$ $= 2.68 \times 10^{-9}$ m $= 2.7 \times 10^{-9}$ m
(iv)	The electrons incident on the double slit have a range of speeds / kinetic energies.
	Hence, electrons have a range of de Broglie wavelengths, producing fringes with different spacings.
	The fringes overlap causing the interference pattern becomes less distinct / blurred.